

Final Project Report

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1. Introduction

This project presents the implementation of a 16-bit MIPS-like pipelined processor using Logisim Evolution. The processor supports R-type, I-type, and J-type instructions and implements a five-stage pipeline architecture. The project objective was to design a complete datapath, control unit, forwarding unit, and hazard detection logic capable of executing instructions correctly while handling data and control hazards.

2. Instruction Set Architecture

The processor implements a custom 16-bit ISA with seven general-purpose registers (R1–R7), while R0 is hardwired to zero. Instructions are divided into three formats:

Instruction Type	Examples	Functionality
R-Type	ADD, SUB, AND, OR, XOR, SLT	Register-to-register arithmetic and logic
I-Type	ADDI, LW, SW, BEQ, BNE	Immediate operations, memory access, branching
J-Type	J, JAL, Lui	Jump and immediate loading instructions

3. Processor Design and Implementation

The processor was implemented hierarchically in Logisim using several subcircuits. The main top-level circuit used in the design is MainDesign. The design includes the following important modules identified in the implementation file:

- Reg16bit – 16-bit register implementation
- ALU – Arithmetic and Logic Unit
- Extender – Sign extension and immediate processing
- NextPC – Next Program Counter calculation logic
- ALU_ctrl – ALU control logic
- MainCtrl – Main control unit
- Forward – Data forwarding and hazard handling logic
- MainCtrlRegs – Pipeline control signal registers

4. Datapath Architecture

The processor datapath follows a standard five-stage pipelined architecture:

1. Instruction Fetch (IF)

2. Instruction Decode (ID)
3. Execute (EX)
4. Memory Access (MEM)
5. Write Back (WB)

Pipeline registers were inserted between each stage to allow multiple instructions to execute simultaneously. The processor uses separate instruction and data memories with 4096-word capacity.

5. Arithmetic and Logic Unit (ALU)

The ALU was designed to support all arithmetic, logical, comparison, and shifting operations required by the project specification.

Operation Group	Supported Operations
Arithmetic	ADD, SUB
Logic	AND, OR, XOR, NOR
Comparison	SLT, SLTU
Shift/Rotate	SLL, SRL, SRA, ROL

6. Hazard Detection and Forwarding

To improve performance and maintain correct execution, the processor implements data forwarding and pipeline stalling mechanisms.

- RAW hazards resolved using forwarding multiplexers
- Load-use hazards handled using one-cycle stalls
- Branch and jump hazards handled using pipeline flushing
- Forwarding paths added from EX, MEM, and WB stages

7. Control Unit

The control unit consists of a Main Control Unit and an ALU Control Unit. The Main Control Unit decodes instruction opcodes and generates the required control signals for datapath execution. The ALU Control Unit generates ALU operation signals based on instruction type and function field.

- RegWrite
- MemRead
- MemWrite
- MemtoReg
- ALUSrc
- Branch
- Jump

- ALUCtrl

8. Simulation and Testing

The processor was tested using the test cases provided on Blackboard in addition to several debugging and verification attempts performed by the team. Most test cases produced the expected correct results during simulation; however, some cases showed unexpected outputs. After extensive debugging, signal tracking, and verification of the datapath and control behavior, it was observed that certain PC values and instruction contents stored in the ROM did not fully match the reference table provided on Blackboard. This may have contributed to the incorrect simulation results observed in some test cases. Despite these issues, significant effort was made to analyze and resolve the problem, and the processor was successfully verified for most of the implemented functionalities.

9. Teamwork and Contribution

All team members worked together collaboratively throughout the entire project. The design, implementation, testing, debugging, and documentation tasks were completed jointly as a team rather than being divided individually. Regular meetings and discussions were conducted to ensure that every member contributed to all stages of the processor development.

10. Conclusion

This project successfully implemented a functional 16-bit pipelined processor using Logisim Evolution. The processor supports multiple instruction formats, pipeline execution, forwarding, and hazard handling mechanisms. The implementation demonstrated important computer architecture concepts including datapath design, pipeline control, forwarding, and stalling.